

Bridging the “real” and “imaginary” worlds

Le Chen
Auburn University

April 19, 2022

Stochastic Analysis Seminar

Auburn University

Abstract: In this talk, we will show how one can use the fractional calculus to connect the equations of distinct natures, or to be more precise, the parabolic equations with the hyperbolic ones. We will present some recent work that demonstrates how properties of equations in these two worlds can be interpolated. This talk is prepared with a general audience in mind, so no knowledge of advanced mathematics, such as stochastic analysis, is required. Graduate students are especially welcome to attend.

Finally, as an advertisement, mathematical modeling using fractional calculus has attracted a lot of attentions recently. One may find more information on the semester long program in the prestigious Issac Newton Institue for Mathematical Sciences at Oxford University:

<https://www.newton.ac.uk/event/fde2/>